

Essential Information

Instructor	Ed Bueler elbueler@alaska.edu
Class meeting	MWF 10:30–11:30 am, Chapman 107
Section CRNs	in-person 901: 35010 online 701: 35018 (zoom link here)
Public website	bueler.github.io/fa
Canvas website	canvas.alaska.edu/courses/30068
Required text	K. Saxe, <i>Beginning Functional Analysis</i>, Undergraduate Texts in Mathematics, Springer 2010

Description

Functional analysis is the theory of infinite-dimensional vector spaces. We will study the topology, subspaces, and linear maps on these vector spaces.

Functional analysis is the mathematical home of partial differential equations, boundary value problems, quantum mechanics, continuous signals, the finite element method, field theories (electromagnetic and gravitational), continuum mechanics, fluids, and elasticity.

The UAF catalog description of the course is this: *Study of Banach and Hilbert spaces, and continuous linear maps between them. Linear functionals and the Hahn-Banach theorem. Applications of the Baire Category theorem. Compact operators, self adjoint operators, and their spectral properties. Weak topology and its applications.*

The above description is accurate, and all of these topics will be covered. However, I will emphasize some aspects and gloss others.

This course will focus on: Spaces of continuous and integrable functions. Hilbert and Banach spaces including Sobolev spaces. Connections with finite-dimensional linear algebra and partial differential equations. Bounded and compact operators (linear maps). Approximation using polynomials, trigonometric functions, and finite-dimensional subspaces. Weak forms and background for the finite element method.

Mathematically-inclined students from the sciences and engineering are encouraged to register, as are graduate students in mathematics looking for an elective with practical relevance. This course is particularly aimed at students interested in the mathematics of partial differential equations, including fluids and elastic solids, and the finite element method. On the other hand, this will definitely be a graduate mathematics course; most problems will be proofs!

Prerequisites

Officially: *MATH F314 Linear Algebra* and *MATH 401 Introduction to Real Analysis*, or permission of instructor. Recommended: *MATH F422 Introduction to Complex Analysis* and *MATH F641 Real Analysis*.

For graduate students with a background from another university, these prerequisites assume a rigorous, though introductory, exposure to the analysis of real functions, adequate linear algebra, and the basics of complex numbers. Quite a bit of mathematical maturity and motivation is assumed.

The difficulty of the assigned work will depend on the student's comfort doing proofs, especially with vector spaces, sequences, real functions, convergence, continuity, and integrals.

The Hybrid Classroom

There are two sections of the class, in-person (901) and online (701). They are treated as one course and occur simultaneously. In this "hybrid" set-up, each lecture will be a Zoom session generated from Chapman 107. (The link for the Zoom session is [in Canvas](#).) It will be recorded, and the recordings will be linked from inside Canvas only; they are not public.

I will try to treat all students the same regarding participation during class time and for proctored assessments.

For online students, please sign into the Zoom session (from Canvas) just before class starts. Please participate as energetically as you can, and, if possible, keep your camera on. Regarding in-class group work, check for worksheet PDFs [on the Daily Log tab of the public site](#) before class starts. You will need to schedule proctoring for the in-class assessments (see below), or attend in-person on those days.

Schedule and Online Materials

The [public course website](#) includes a [day-by-day schedule](#) listing the textbook sections to be covered, the due date of each homework Assignment, and timing of the Exams. Please consult this schedule frequently. It is subject to change, but it will be kept up to date.

Most course materials (syllabus, schedule, homework Assignments, etc.) will be posted on the [public website](#). Some private-access course materials (zoom link, grades, solutions) will go on the [Canvas site](#).

Office Hours and Communication

My Office Hours are online at bueler.github.io/OffHrs.htm; I'm in Chapman 306C. Students can also schedule meetings with me outside of regular office hours; please send an email.

I will use Canvas to send announcements. If I need to contact you outside of class times, I'll try to email via Canvas. (Please set your email address in Canvas to one that you check regularly!)

Evaluation and Grades

Homework	nearly weekly	40%
Midterm Exam 1	in-class Monday 2 March	20%
Midterm Exam 2	in-class Monday 13 April	20%
Final Exam	in-class Friday 1 May, 10:15am–12:15pm	20%
total		100%

Scores will be summed and the course grade will be assigned as follows:

A	93–100%	B-	79–81%	D+	65–67%
A-	90–92%	C+	76–78%	D	60–64%
B+	87–89%	C	68–75%	D-	57–59%
B	82–86%	C-	not given	F	$\leq 56\%$

These ranges are a guarantee and a lower bound. I reserve the right to increase your grade above these ranges based on the actual difficulty of the work and/or on average class performance. Any such increases will preserve grade ordering by weighted total score.

Homework

Homework Assignments and their due dates will be posted at the [public website](#). Homework consists of by-hand computations, rigorously-justified examples and counter-examples, and proofs.

Homework is due at the start of class. Late homework is not accepted. If you have unavoidable circumstances which do not allow you to turn in an Assignment on time then please contact me (elbueler@alaska.edu) in advance.

Please turn in homework assignments electronically, as clear and well-ordered PDFs, at the corresponding Assignment on the [Canvas site](#).

You may talk to other students about the homework, and you may use internet and generative AI resources in your solutions. However, actual comprehension of the ideas on the Homework is required to get a passing grade in the course! The in-class and on-paper Exams are genuine assessments which will allow no technology whatsoever. These assessments will have a large overlap with homework content, and some questions from homework will be duplicated onto the Exams, so the way to succeed in the course is to take the homework seriously.

Exams

There will be two in-class, hour-long Midterm Exams covering definitions and problems based on core concepts. Problems will be very similar to, or shortened versions of, already-assigned homework problems. The in-class Final Exam will require you to be familiar with major concepts we have studied. I will describe the format of this Exam in more detail later.

Make-up Exams will be given only for documented extenuating circumstances, at my discretion. Department policy (below) does not allow me to move the time of the Final Exam.

Department of Mathematics & Statistics Rules and Policies**Incomplete Grade**

Incomplete (I) will only be given in DMS courses in cases where the student has completed the majority (normally all but the last three weeks) of a course with a grade of C or better, but for personal reasons beyond his/her control has been unable to complete the course during the regular term. Negligence or indifference are not acceptable reasons for granting an incomplete grade.

Late Withdrawals

A withdrawal after the deadline from a DMS course will normally be granted only in cases where the student is performing satisfactorily (i.e., C or better) in a course, but has exceptional reasons, beyond his/her control, for being unable to complete the course. These exceptional reasons should be detailed in writing to the instructor, Department Chair and the Dean.

No Early Final Examinations

Final examinations for DMS courses shall not be held earlier than the date and time published in the official term schedule. Normally, a student will not be allowed to take a final exam early. Exceptions can be made by individual instructors, but should only be allowed in exceptional circumstances and in a manner which doesn't endanger the security of the exam.

Academic Dishonesty

Academic dishonesty, including cheating and plagiarism, will not be tolerated. It is a violation of the Student Code of Conduct and will be punished according to UAF procedures.

Student protections and service statement

Every qualified student is welcome in my classroom. As needed, I am happy to work with you, Disability Services, Veterans' Services, Rural Student Services, and so on, to find reasonable accommodations. Students at this University are protected against sexual harassment and discrimination (Title IX), and minors have additional protections. For more information on your rights as a student and the resources available to you to resolve problems, please go the following site: www.uaf.edu/handbook.

Official UAF Syllabus Addendum**Student protections statement**

The university respects and upholds the principles of due process and a fair and equitable process as specified in the Board of Regents' Policy 09.02 Student Rights and Responsibilities. For more information regarding the rights and responsibilities of students, refer to the Office of Rights, Compliance and Accountability website. You are encouraged to read the

Board of Regents' policy carefully to fully understand your responsibilities to our community.

We strive to create a safe and respectful environment for all members of our community. If you have questions about expectations of you as a student or believe your rights are being violated, we encourage you to reach out to the Office of Rights, Compliance and Accountability for help. UAF reserves the right to suspend, expel or take other necessary and appropriate action in cases where a student is unable or unwilling to uphold community standards and campus safety.

For more information on your rights as a student and the resources available to you to resolve problems, please go to the following site: catalog.uaf.edu/academics-regulations/students-rights-responsibilities

Disability services statement

I will work with the Office of Disability Services to provide reasonable accommodation to students with disabilities.

ASUAF advocacy statement

The Associated Students of the University of Alaska Fairbanks, the student government of UAF, offers advocacy services to students who feel they are facing issues with staff, faculty, and/or other students specifically if these issues are hindering the ability of the student to succeed in their academics or go about their lives at the university. Students who wish to utilize these services can contact the Student Advocacy Director by visiting the ASUAF office or emailing asuaf.office@alaska.edu.

Student Academic Support

- Communication Center (907-474-7007, uaf-commcenter@alaska.edu, Student Success Center, 6th Floor Room 677 Rasmuson Library)
- Writing Center (907-474-5314, uaf-writing-center@alaska.edu, Student Success Center, 6th Floor Room 677 Rasmuson Library)
- UAF Math Services (907-474-7332, uaf-traccloud@alaska.edu)
 - Drop-in tutoring, Student Success Center, 6th Floor Room 672 Rasmuson Library
 - 1:1 tutoring (by appointment only), 6th Floor Room 677 Rasmuson Library
 - Online tutoring (by appointment only), available at the Student Success Center www.uaf.edu/dms/mathlab
- Developmental Math Lab, Gruening 406
- The Debbie Moses Learning Center at CTC (907-455-2860, 604 Barnette St, Room 120, www.ctc.uaf.edu/student-services/student-success-center)

- For more information and resources, please see the Academic Advising Resource List (www.uaf.edu/advising/students/index.php)

Student Resources

- Disability Services (907-474-5655, uaf-disability-services@alaska.edu, 110 Eielson Building)
- Student Health & Counseling, free counseling sessions available (907-474-7043, www.uaf.edu/chc/appointments.php, Whitaker Building, Room 206, Health, Safety & Security Bldg — same building as Fire and Police)
- Office of Rights, Compliance and Accountability (907-474-7300, uaf-orca@alaska.edu, 3rd Floor, Constitution Hall)
- Associated Students of the University of Alaska Fairbanks (ASUAF) or ASUAF Student Government (907-474-7355, asuaf.office@alaska.edu, Wood Center 119)

Nondiscrimination statement

Nondiscrimination statement: The University of Alaska is an equal opportunity/equal access employer, educational institution and provider. The University of Alaska does not discriminate on the basis of race, religion, color, national origin, citizenship, age, sex, physical or mental disability, status as a protected veteran, marital status, changes in marital status, pregnancy, childbirth or related medical conditions, parenthood, sexual orientation, gender identity, political affiliation or belief, genetic information, or other legally protected status. The University's commitment to nondiscrimination, including against sex discrimination, applies to students, employees, and applicants for admission and employment. Contact information, applicable laws, and complaint procedures are included on UA's statement of nondiscrimination available at www.alaska.edu/nondiscrimination.

UAF Office of Rights, Compliance and Accountability
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